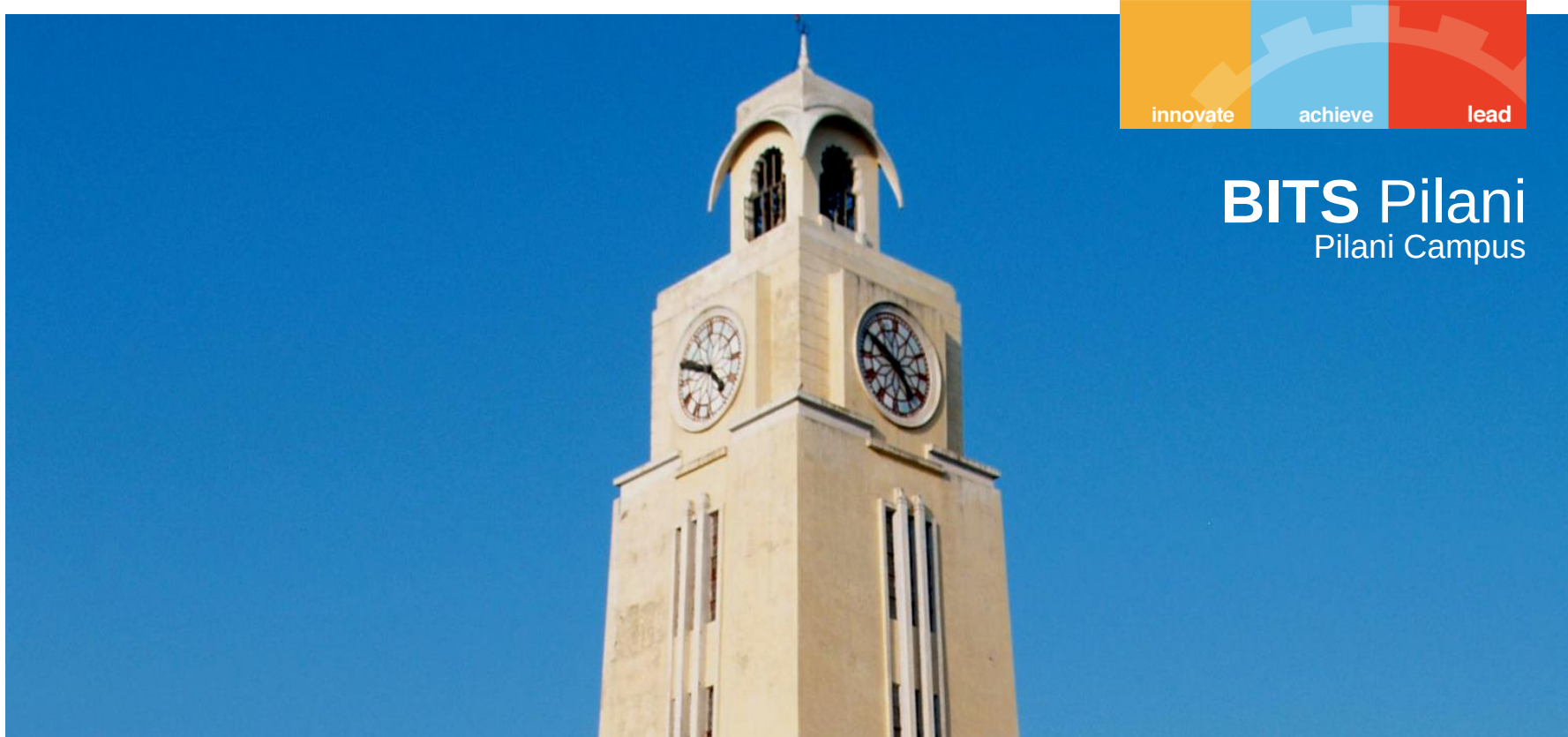




BITS Pilani
Pilani Campus



CHEM F111 : General Chemistry

Semester II: AY 2023-24

Lecture - 10, 5th February 2024, Monday

Last class:

- Total wavefunction for the $2p_z$ orbital; how the angular part decides the shape of the orbital; No imaginary part in the original solution of SE for $2p_z = 2p_0$;
- $2p_x$ and $2p_y$ as the linear combination of $2p_{+1}$ and $2p_{-1}$ orbitals; these combinations still will be eigen functions of energy and angular momentum operators;
- Similar explanations for the five d orbitals; only d_{z^2} is real and original; other 4 are linear combinations of original solutions of SE.
- Spectral transitions; selection rules for hydrogenic atom transitions;
 $\Delta l = \pm 1$; $\Delta m_l = 0, \pm 1$;
- Fermions, Bosons; antisymmetric principle; Pauli exclusion principle

- Schrödinger Equation for a many-electron atom is highly complex because all electrons are interacting with each other.
- Helium Atom (Two electrons): electrons 1 and 2;
- Mathematical expression for orbitals and energies cannot be given.
- Orbital approximation: The wavefunction of non-interacting particles is the product of wavefunction of each particle.
$$\psi(1,2,\dots) = \psi_1(1)\psi_2(2)\dots$$
- The orbital approximation allows us to express the electronic structure of an atom in the form of configuration: a statement of orbital representation that are occupied. (*but not necessarily in the ground state*)
H (Electronic Configuration): $1s^1$

Orbital approximation – He atom

- We construct a helium wave function as the product of hydrogenic orbitals with $Z = 2$.

$$\psi(1,2) = \phi(1)\phi(2)$$

- Each orbital may be thought of as being hydrogen-like with an effective nuclear charge.
- The 1st electron of helium occupies the 1s hydrogenic orbital and since Z for He is 2 so this 1s orbital can be assumed to be more compact that it can accommodate a second electron leading to He (Electronic Configuration): $1s^2$
- If one disregards the inter-electronic repulsion, the ground state wavefunction of He may be written as (electron 1 is at radius r_1 and electron 2 is at radius r_2)

$$\psi(1,2) = (8/\pi a_0^3)^{1/2} e^{-2r_1/a_0} (8/\pi a_0^3)^{1/2} e^{-2r_2/a_0}$$

What will be the form of the wavefunction?

$$\psi(1,2) = \phi(1)\phi(2) \alpha(1)\alpha(2)$$

Order of multiplying the functions does not change the value of the product

$$\psi(1,2) = \phi(1)\phi(2) \beta(1)\beta(2)$$

An electron is fermion and fermions' wave function must be antisymmetric with respect to interchange: **Fermions** $\psi(2,1) = -\psi(1,2)$

$$\psi(1,2) = 2^{-1/2} \phi(1)\phi(2) \{ \alpha(1)\beta(2) + \beta(1)\alpha(2) \}$$

$$\psi(1,2) = 2^{-1/2} \phi(1)\phi(2) \{ \alpha(1)\beta(2) - \beta(1)\alpha(2) \}$$

It should be antisymmetric with respect to exchange of the electrons' coordinates and should be indistinguishable.

$$\Psi(1,2) = 2^{-1/2} \phi(1)\phi(2) \{ \alpha(1)\beta(2) - \beta(1)\alpha(2) \}$$



- **Shielding & Penetration of Orbitals**
- **Order of occupation** : Electronic config. of atoms, cations and Anions

Rule 1. Order of occupation of atomic orbitals follows:

1s 2s 2p 3s 3p **4s 3d** 4p 5s 4d 5p 6s 5d 4f 6p... (*Aufbau principle*)

Rule 2. Each orbital may accommodate up to two electrons
(*Pauli principle*)

Rule 3. Electrons occupy different orbitals of a given subshell before doubly occupying any one of them.

Rule 4. In its ground state, atom adopts configuration with greatest number of unpaired electrons (*Hund's rule*)

- **Periodic Properties**

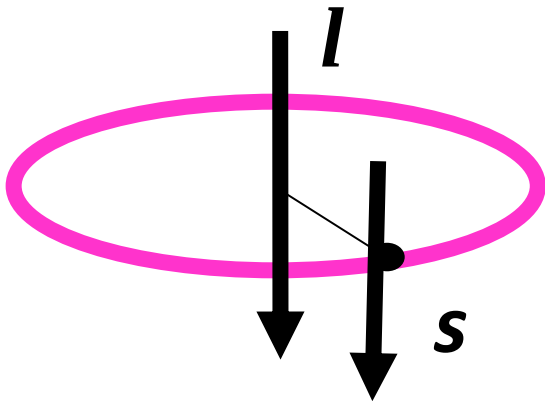
Atomic Radius, Ionization Energy, Electron affinity

- The spectra of many-electron atoms can be complicated.
- For multi-electron atoms, the spectrum or transitions between different energy values could only be explained if we define a new system of **term/energy levels/states** that takes into account of the interaction between electrons (in terms of considering the coupling between orbital and spin angular momentum of electrons).

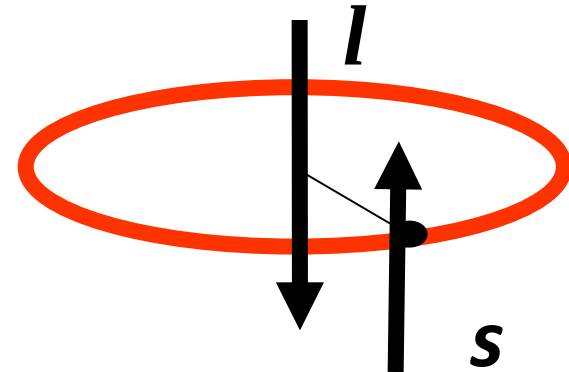
Spin-orbit interaction, j



- Electrons have both spin angular momentum and orbital angular momentum ($l > 0$)
- Moving charges generate magnetic fields : Magnetic moment from spin & Magnetic moment from orbital angular momentum interact.



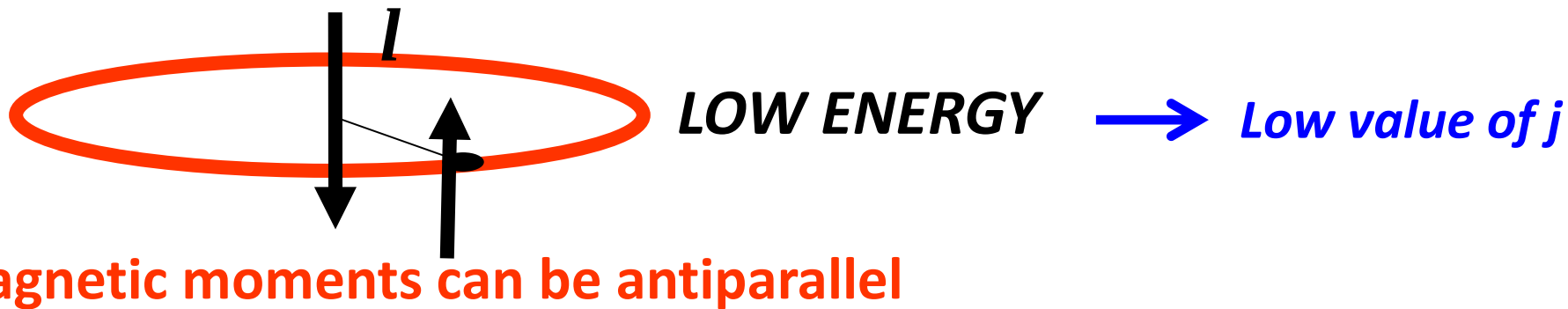
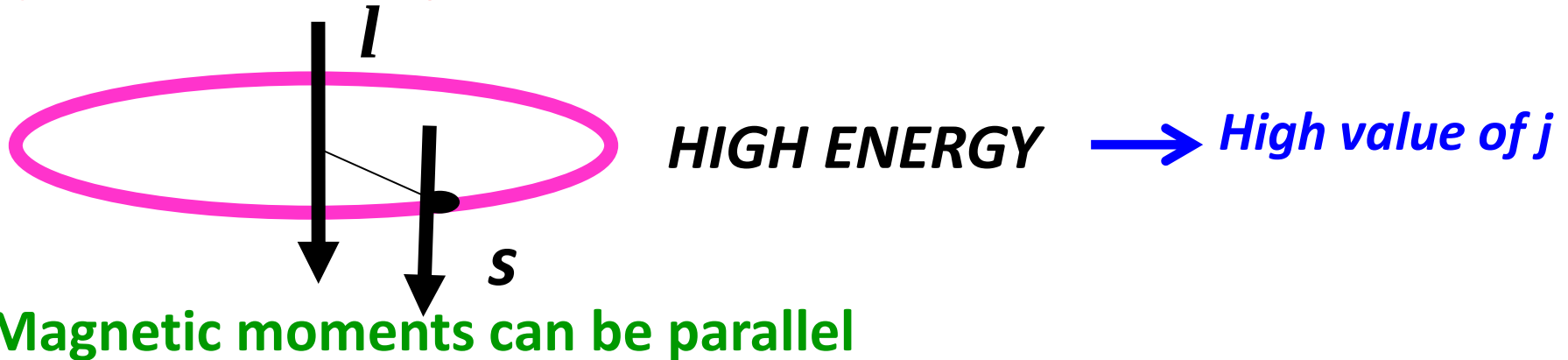
Magnetic moments can be parallel
(angular momenta are parallel)



Magnetic moments can be
antiparallel (angular
momenta are antiparallel)

Spin-orbit interaction, j

- Depending upon the relative orientation of spin- and orbital- angular momenta, the total angular momentum can be calculated in terms of a quantum number, j



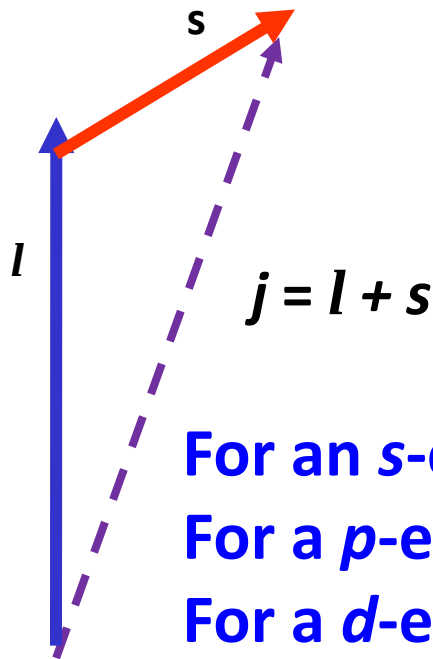
- What are the allowed values of j ?

Spin-orbit interaction of a single electron



- Relative orientation of the spin and orbital angular momenta.
- Use of vector representation to note coupling of spin and orbital angular momenta

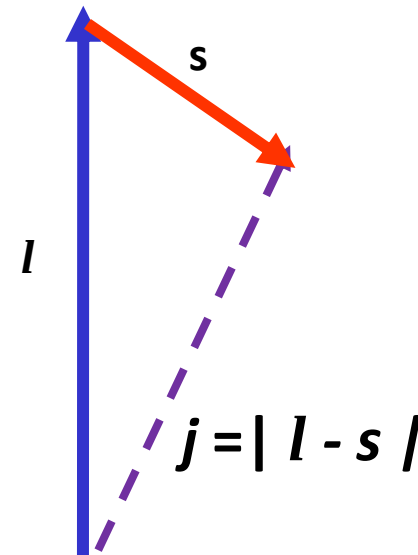
Two angular momenta
adding



Total angular
momenta could be
represented

$$j = |l \pm s|$$

Two angular momenta
opposing



For an s-electron: $l = 0, s = 1/2; j = 1/2$ only

For a p-electron: $l = 1, s = 1/2; j = 3/2$ or $1/2$

For a d-electron: $l = 2, s = 1/2; j = 5/2$ or $3/2$

For two electron system: total orbital & spin angular momentum



Total orbital angular momentum could be represented as

$$L = (l_1 + l_2), (l_1 + l_2 - 1), (l_1 + l_2 - 2), \dots, |l_1 - l_2|$$

Two orbital ang.
momenta are in
same direction

Two orbital ang.
momenta are in
opposite direction

Clebsch-Gordan Series

Similarly, total spin angular
momentum could be
represented as

$$S = (s_1 + s_2), (s_1 + s_2 - 1), (s_1 + s_2 - 2), \dots, |s_1 - s_2|$$

Total angular momentum



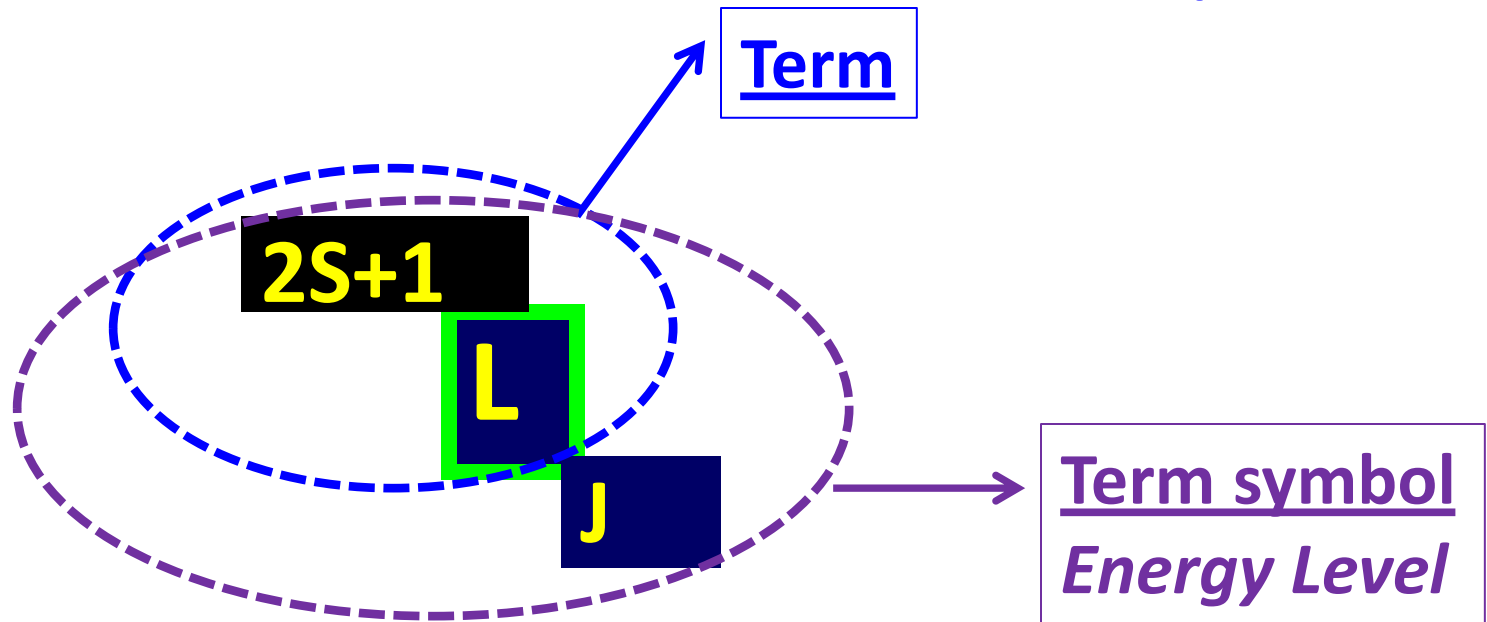
We have seen that j gives relative orientation of spin and orbital angular momentum of a single electron.

Total angular momentum quantum number J does the same thing for multi-electron system

$$J = L+S, L+S - 1, \dots, |L-S|$$

- This scheme to arrive at the total angular momentum is called **Russell-Saunders coupling**. (RS coupling or LS coupling)
- It is based on the fact that total orbital and total spin angular momentum of an electron couples (spin-orbit coupling) to give resultant momenta and all these resultants couple to give an overall total angular momentum.

- An electronic configuration is associated with a set of terms, which are associated with a set of levels, and each level is further associated with a set of electronic states for that atom/ion.



- The core or inner electrons do not contribute to the overall angular momentum of the atom. We need to consider only the electrons of the unfilled shell.

Information about S and
multiplicity

2S+1

Information about L,
Total orbital angular
momentum quantum
number

L

J

Code for conversion
to letter:

L = 0(**S**), 1(**P**), 2(**D**),
3(**F**), 4(**G**), 5(**H**), 6(**I**),...

Information about J, Total
angular momentum (spin and
orbital) quantum number

Electronic state for a s^1 configuration



Configuration: $1s^1$

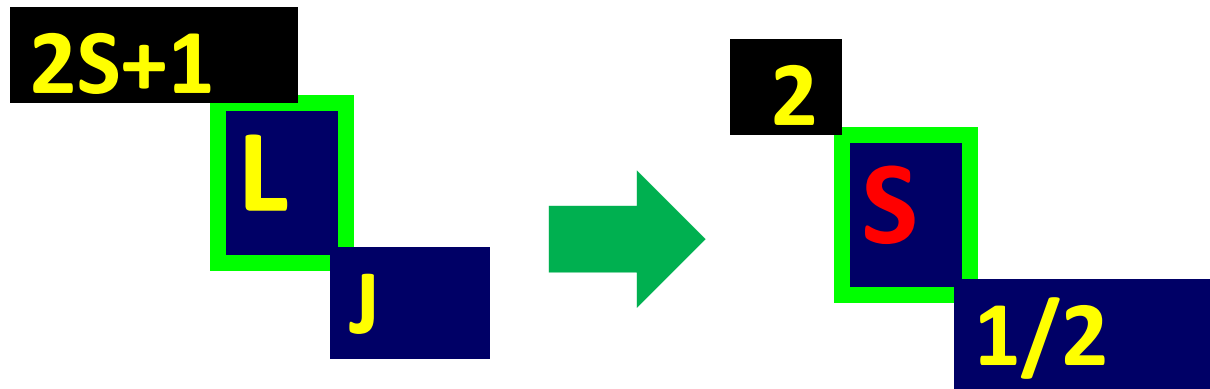
$$l_1 = 0 = L$$

$$s_1 = 1/2 = S$$

$$j = J = 0 + 1/2, \dots, |0 - 1/2|$$

$$J = 1/2$$

The state for this configuration is represented by:



Here, J will have two different orientations given by $m_j = 1/2$ and $m_j = -1/2$ which are only allowed.

Electronic states for a $2p^1$ system



EC: $1s^2 2s^2 2p^1$

Completely filled subshells do not contribute to the total angular momentum of an atom

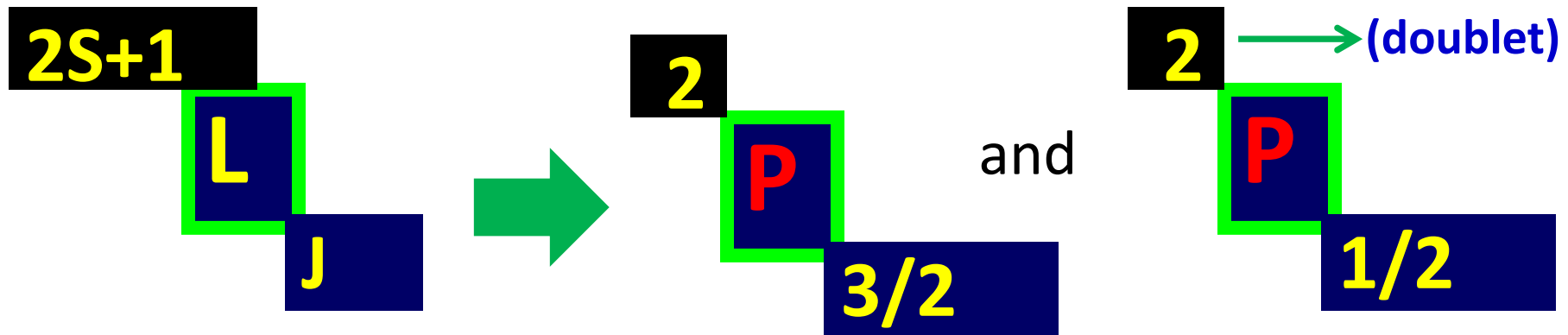
$$l_1 = 1 = L$$

$$s_1 = 1/2 = S$$

$$J = 1 + 1/2, \dots, |1 - 1/2|$$

$$J = 3/2, 1/2$$

The states for this configuration are represented by:

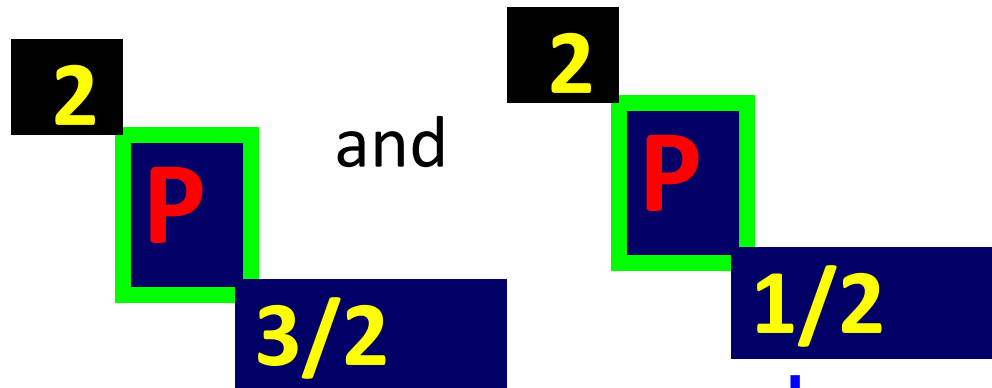


For each j , there will be $(2j+1)$ possible states/different orientations of j given by m_j : $-j$ to $+j$ (in integral steps)

Electronic states for a $2p^1$ system

EC: $1s^2 2s^2 2p^1$

The states for this configuration is represented by two levels represented by term symbols:



For a given configuration,
No. of states = $b! / ((b-a)! a!)$
= $6! / ((6-1)! 1!) = 6$ states

b: total possible positions:
a : total no of electron

$(2J+1)$ states in this level = $(2 \times (1/2) + 1) = 2$ degenerate states

$(2J+1)$ states in this level = $(2 \times (3/2) + 1) = 4$ degenerate states

For a given term, number of states $(2L + 1)(2S + 1)$:
only one term: 2P Term: $(2 \times 1 + 1)(2 \times 1/2 + 1) = 6$ states

Term symbol: summary



- A term is a combination of the one of the allowed total orbital angular momentum values (L_i) with one of the allowed total spin angular momentum values (S_i);
- Each term will have a number of levels; the number of levels is decided by the number of possible combinations of the L_i and S_i (specified by the term) to get total angular momentum values (J_i).
- A term symbol consists of all these three values L_i , S_i , and J_i .
- Each level will have a number states of same energy; for a given J , the number of states in a level is given by $2J + 1$, with m_j values from $-J$ to $+J$ with increment by one step.

Term symbol: two p electrons (in different shells)



• A term is a combination of the one of the allowed **total orbital angular momentum** values (L_i) with one of the allowed **total spin angular momentum** values (S_i);

Ex:

One of the excited state configuration of C is $[\text{He}]2s^2\mathbf{2p^13p^1}$
*Allowed **total orbital angular momentum** values*

$$l_1 = 1, l_2 = 1$$

$$L = l_1 + l_2, l_1 + l_2 - 1, l_1 + l_2 - 2, \dots, |l_1 - l_2|.$$

$$L = (1+1=2), (1+1-1=1), |1-1=0|.$$

L = 2(D), 1 (P), 0 (S) DIFFERENT POSSIBLE RELATIVE ORIENTATIONS OF THE RESULTANT ORBITAL ANGULAR MOMENTA OF A 2-ELECTRON SYSTEM

Term symbol: two p electrons (in different shells)



• A term is a combination of the one of the allowed **total orbital angular momentum** values (L_i) with one of the allowed **total spin angular momentum** values (S_i);

Ex:

One of the excited state configuration of C is $[\text{He}]2s^2\mathbf{2p^13p^1}$

*Allowed **total spin angular momentum** values*

$$S = s_1 + s_2, s_1 + s_2 - 1, s_1 + s_2 - 2, \dots, |s_1 - s_2|.$$

$$s_1 = \frac{1}{2} \quad \& \quad s_2 = \frac{1}{2}$$

$$S = 1, 0$$

Term symbol: two p electrons (in different shells)



• A term is a combination of the one of the allowed total orbital angular momentum values (L_i) with one of the allowed total spin angular momentum values (S_i);

Ex:

One of the excited state configuration of C is $[\text{He}]2s^2 2p^1 3p^1$

A term is a combination of the one of the allowed L_i and S_i

$L = 2(D), 1(P), 0(S)$

$S = 1, 0$

$2S+1$

L

J

Triplet terms: $^3D, ^3P, ^3S$

Singlet terms: $^1D, ^1P, ^1S$

Term symbol: two p electrons (in different shells)



- Each term will have a number of levels; the number of levels is decided by the number of possible combinations of the L_i and S_i (specified by the term) to get total angular momentum values (J_i).

Ex:

$L = 2(D), 1(P), 0(S)$

$S = 1, 0$

$2S+1$

L

J

Triplet terms: $^3D, ^3P, ^3S$

Singlet terms: $^1D, ^1P, ^1S$

$$J = L+S, L+S-1, \dots, |L-S|$$

For each term J needs to be calculated separately

For 3D , the levels are $^3D_3, ^3D_2, ^3D_1$ as $J = 3, 2, 1$ [$L=2, S=1$]

For 1D , the level is 1D_2 as $J = 2$ [$L=2, S=0$]

Term symbol: two p electrons (in different shells)



3D term: $L = 2$ and $S = 1$, So $J = 3, 2, 1$. Levels are

3D_3

3D_2

3D_1

3P term: $L = 1$ and $S = 1$, So, $J = 2, 1, 0$. Levels are

3P_2

3P_1

3P_0

3S , : $L = 0$ and $S = 1$, $J = 1$. Level is

3S_1

1D term: $L = 2$ and $S = 0$, $J = 2$. Level is: 1D_2

1P term: $L = 1$ and $S = 0$, $J = 1$. Level is: 1P_1

1S term: $L = 0$ and $S = 0$, $J = 0$. Level is: 1S_0

Term symbol: two p electrons (in different shells)



- Each level will have a number states of same energy; for a given J , the number of states in a level is given by $2J + 1$, with m_J values from $-J$ to $+J$ with increment by one step.

$$\begin{aligned} {}^3D_3 &\longrightarrow 2J + 1 = 7 \text{ degenerate states} \\ {}^3D_2 &\longrightarrow 2J + 1 = 5 \text{ degenerate states} \\ {}^3D_1 &\longrightarrow 2J + 1 = 3 \text{ degenerate states} \end{aligned}$$

Term symbol: two p electrons (in different shells)



3D term: $L = 2$ and $S = 1$, So $J = 3, 2, 1$. Levels are

$^3D_3 \rightarrow 7$ degenerate states

$^3D_2 \rightarrow 5$ degenerate states

$^3D_1 \rightarrow 3$ degenerate states

3P term: $L = 1$ and $S = 1$, So, $J = 2, 1, 0$. Levels are

$^3P_2 \rightarrow 5$ degenerate states

$^3P_1 \rightarrow 3$ degenerate states

$^3P_0 \rightarrow 1$ non-degenerate

3S , : $L = 0$ and $S = 1$, $J = 1$. Level is

$^3S_1 \rightarrow 3$ degenerate states

1D term: $L = 2$ and $S = 0$, $J = 2$. Level is: $^1D_2 \rightarrow 5$ degenerate states

1P term: $L = 1$ and $S = 0$, $J = 1$. Level is: $^1P_1 \rightarrow 3$ degenerate states

1S term: $L = 0$ and $S = 0$, $J = 0$. Level is: $^1S_0 \rightarrow 1$ non-degenerate

Term symbol: two p electrons (in different shells)



3D term: $L = 2$ and $S = 1$, So $J = 3, 2, 1$. Levels are

$$\begin{array}{lcl} ^3D_3 & \longrightarrow & 7 \text{ degenerate states} \\ ^3D_2 & \longrightarrow & 5 \text{ degenerate states} \\ ^3D_1 & \longrightarrow & 3 \text{ degenerate states} \end{array} \left. \vphantom{\begin{array}{l} ^3D_3 \\ ^3D_2 \\ ^3D_1 \end{array}} \right\} \begin{array}{l} 15 \text{ states} \\ (2L + 1)(2S + 1) \\ (2 \times 2 + 1)(2 \times 1 + 1) = 15 \end{array}$$

3P term: $L = 1$ and $S = 1$, So, $J = 2, 1, 0$. Levels are

$$\begin{array}{lcl} ^3P_2 & \longrightarrow & 5 \text{ degenerate states} \\ ^3P_1 & \longrightarrow & 3 \text{ degenerate states} \\ ^3P_0 & \longrightarrow & 1 \text{ non-degenerate} \end{array} \left. \vphantom{\begin{array}{l} ^3P_2 \\ ^3P_1 \\ ^3P_0 \end{array}} \right\} \begin{array}{l} 9 \text{ states} \\ (2L + 1)(2S + 1) \\ (2 \times 1 + 1)(2 \times 1 + 1) = 9 \end{array}$$

3S term, : $L = 0$ and $S = 1$, $J = 1$. Level is

$$^3S_1 \longrightarrow 3 \text{ degenerate states} \left. \vphantom{^3S_1} \right\} \begin{array}{l} 3 \text{ states} \\ (2 \times 0 + 1)(2 \times 1 + 1) = 3 \end{array}$$

1D term: $L = 2$ and $S = 0$, $J = 2$. Level is: $^1D_2 \longrightarrow 5 \text{ deg. states}$ } 5 states

1P term: $L = 1$ and $S = 0$, $J = 1$. Level is: $^1P_1 \longrightarrow 3 \text{ deg. states}$ } 3 states

1S term: $L = 0$ and $S = 0$, $J = 0$. Level is: $^1S_0 \longrightarrow 1 \text{ non-deg.}$ } 1 states

Term symbol: two p electrons (in different shells)



For a given term, number of states $(2L + 1)(2S + 1)$

3D term, 15

3P term: 9

3S term: 3

1D term: 5

1P term: 3

1S term: 1

Total no. of states: 36

For a given conf., number of states, $b!/(b-a)!a!$

b: total possible positions

a : total no of electron

For a $2p^13p^1$

number of states =

$$6!/(6-1!)1! \times 6!/(6-1!)1!$$

number of states = 36

$^3D, ^3P, ^3S, ^1D, ^1P, ^1S$: All levels exist for this configuration

Determining the ground state : Hund's rule



I The term with the greatest spin multiplicity lies lowest in energy. The ground state therefore possesses the most unpaired spins because this gives the minimum electrostatic repulsion.

So one of the terms 3D , 3P , 3S arising from $[\text{He}]2s^22p^13p^1$ configuration lies lowest. $[^3D, ^3P, ^3S, ^1D, ^1P, ^1S]$

II The term with the greatest orbital angular momentum lies lowest in energy.

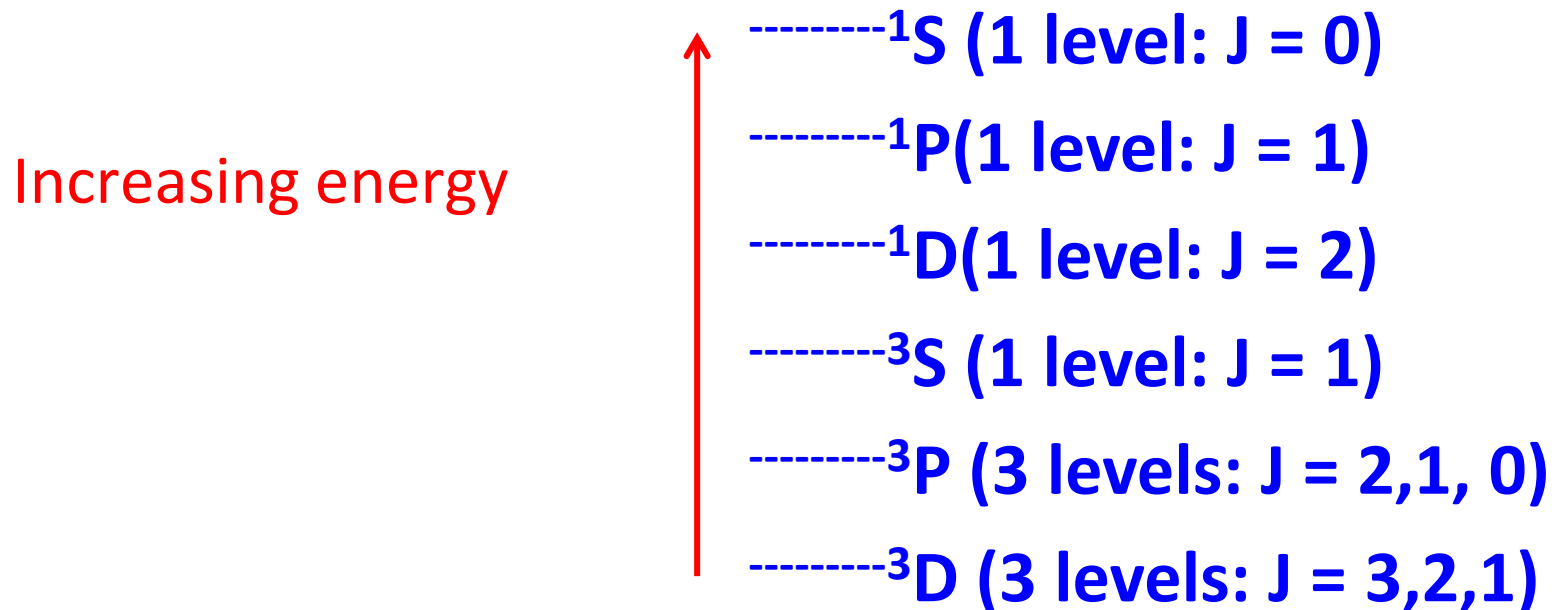
So, 3D term mentioned above will lie lowest of all the terms.

IIIa The level with lowest J lies lowest in energy if the subshell is less than half filled.

IIIb The level with highest J lies lowest in energy if the subshell is more than half filled.

Rule III is not applicable for excited configurations.

Excited configurations: Rule I & Rule II (Hund's rule) can decide the order of terms.



Excited configuration: Rule III (Hund's rule is not applicable)



Point I. We can determine the energy levels/terms/electronic states for a configuration [GS/ES] and arrange them in increasing order of energy.

Simple examples in course: $[s^1]$, $[p^1]$, $[d^1]$, $[s^1 p^1]$, $[p^1 p^1]$, $[p^1 d^1]$, etc..

Point II. Without determining all the levels/states, we can directly determine the ground state term symbol for any configuration.

Point III. We can explain experimental spectra using term symbols and selection rules (see the last slide).

- Without determining all the levels/states, one can directly determine the ground state term symbol for any configuration.

Ground state term should correspond to a term with the greatest multiplicity (lowest energy) or a state with most unpaired spins.

For each configuration: we could determine the ground state term first by choosing possible max. S (considering m_s) and then determining the corresponding max. L possible (by considering the m_l of the orbitals).

	m_l		
	+1	0	-1
m_s			

	m_l				
	+2	+1	0	-1	-2
m_s					

Ground state term symbol determination

2p¹ 3p¹ configuration:

For max. S; $s_1 = \frac{1}{2}$ & $s_2 = \frac{1}{2}$

Both electrons are
in parallel spin

$$\text{Max. } S = 1$$

$$= 2S + 1 = 3$$

2 (D) = Max. L possible

$$M_L = (1 + 1)$$

2p ¹	m_l		
	+1	0	-1
m_s			

3p ¹	m_l		
	+1	0	-1
m_s			

Ground state term: ³D

Ground state term symbol determination

2p² configuration:

For max. S; $s_1 = \frac{1}{2}$ & $s_2 = \frac{1}{2}$

Both electrons are
in parallel spin

$$\text{Max. } S = 1$$

$$= 2S + 1 = 3$$

1 (P) = Max. L possible

$$M_L = (1 + 0)$$

	m_l		
2p ²	+1	0	-1
m_s			

Ground state term: **3P**

L = 1 and S = 1,

So J = 2, 1, 0

3P₂, **3P**₁, **3P**₀

G.S term of p⁴ Configuration will be same as p² configuration,
assuming the filling of holes instead of electrons

Ground state term symbol determination

d³ / d⁷ Configuration: **three electrons should be in parallel spin**

For max. S $s_1 = \frac{1}{2}$
 $s_2 = \frac{1}{2}$
 $s_3 = \frac{1}{2}$

Max. S = $\frac{3}{2}$
 $= 2S + 1 = 4$

3 (F) = Max. L possible

$M_L = 2 + 1 + 0$

	m_l				
	+2	+1	0	-1	+2
m_s					

Ground state term: **⁴F**

For a **⁴F** term, J can be calculated
 $L = 3$ and $S = \frac{3}{2}$,
 So, $J = \frac{9}{2}, \frac{7}{2}, \frac{5}{2}, \frac{3}{2}$.

⁴F_{9/2}

⁴F_{7/2}

⁴F_{5/2}

⁴F_{3/2}

Ground state energy level (d⁷ conf.)

Ground state energy level (d³ conf.)

- Without enumerating all the states, what is the term of lowest energy that arises from the ground electronic configuration of N?

- N; p^3

Max. $S = 3/2$

goes only with $M_L = 0$ ($S = 3/2$ with only $L = 0$ (S))

Ground Term: $^4S \longrightarrow ^4S_{3/2}$

d^1 Configuration: $^2D_{3/2}$

d^9 Configuration: $^2D_{5/2}$

d^2 Configuration: 3F_2

d^8 Configuration: 3F_4

p^1 Configuration: $^2P_{1/2}$

p^5 Configuration: $^2P_{3/2}$

Selection rules

For spectroscopic transitions in many-electron atoms:

$$\Delta L = 0, \pm 1 \quad \Delta I = \pm 1$$

$$\Delta S = 0$$

$$\Delta J = 0, \pm 1 \text{ (but } J = 0 \leftrightarrow J = 0 \text{ is forbidden)}$$

For heavier atoms ($Z > 56$), first coupling of l and s for individual electrons occur to get j values for each electron which then couple with other j to get final J

j-j coupling method. (Not in syllabus)

Question: The Sodium D emission line spectra is actually a doublet, i.e., two very closely spaced lines. Explain.

Answer: This means transition from excited conf. of Na ($3p^1$) to ground level conf. of Sodium ($3s^1$)

Due to spin-orbit coupling the $3p^1$ ($l = 1, s = \frac{1}{2}$) configuration actually splits into two levels, corresponding to total angular momentum quantum number $j = J = \frac{3}{2}$, and $\frac{1}{2}$:

$^2P_{3/2}$ & $^2P_{1/2}$

$3s^1$ ($l = 0, s = \frac{1}{2}$) configuration corresponds to total angular momentum quantum number $j = J = \frac{1}{2}$ ($l + s$): $^2S_{1/2}$

Two lines are due to transition:

$^2P_{3/2}$	$\longrightarrow$	$^2S_{1/2}$	$\Delta L = 1, \Delta S = 0, \Delta J = 1$
$^2P_{1/2}$	$\longrightarrow$	$^2S_{1/2}$	$\Delta L = 1, \Delta S = 0, \Delta J = 0$